

INSTITUTE OF ACTUARIES OF INDIA

Subject SP7 - General Insurance Reserving and Capital Modelling

November 2024 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

- 1) It is not stated whether the paid or incurred Bornhuetter- Ferguson method is intended and which of the methods (paid/incurred) chain ladder method it is going to replace.
- 2) Paid B-F method would be an improvement over the chain-ladder method, if a sensible initial expected loss estimate is available.
- 3) This is because its development will be more stable and less affected by early swings in paid amounts.
- 4) However, this method ignores the amount of outstanding reserves set up in respect of reported claims.
- 5) The o/s reserves are normally a valuable source of information, even though in this case it is noted that the reserves are often badly inaccurate.
- 6) When the incurred claims are overstated/more than 100% of the ultimate amounts as it is possible in this case, then the ultimate's may vary significantly of initial expected loss. Thus, if a reduction in incurred claims are expected or there is negative development due to larger O/S reserve, then incurred B-F method is not appropriate.
- 7) Even if there is further modest increase expected in incurred claims, the minimal weight on initial expected losses will not give enough opportunity to exercise a stable influence on the reserves.
- 8) The actuary can consider the results of B-F methods in addition to current methods the company uses.

[Max 5 Marks]

Solution 2:**a)**

Option 1: run – off existing liabilities

Under this option, the company will themselves settle all the outstanding liability.

Advantages

1. Transferring the liabilities could be an expensive option and thus company could save some money.
2. Since this book forms a significant proportion and is long tail, huge liquidity would be required in case the liabilities are to be transferred. Therefore, in this scenario no need to realize/sell off any assets and incur losses due to premature withdrawal.
3. No Immediate tax impact as there is no immediate transfer of liabilities or realization of assets.
4. Easier to administer as no regulatory approvals/documentation is required.

Disadvantages

1. The risk of adverse development of reserves stays with the company.
2. The liabilities might increase due to higher than expected inflation in claims.
3. The investment returns could be lower than expected and thus may result in more losses.
4. Longer tail might lead to deployment of specialist claim personnel for longer duration and thus could be costly.
5. Risk of latent claims can emerge later.

(Max 4)

Option 2: sell the run-off portfolio to another insurer/reinsurer

In this scenario, the liability for the particular line of business is passed entirely to another insurer/reinsurer.

Advantages

1. The original insurer no longer has any remaining exposures, including any subsequent reserve deteriorations.

2. All uncertainties with regards to change in external environment has been passed to another insurer/reinsurer.
3. Lower capital requirement and capital will be freed for other purposes.
4. Quick & fast method of exiting a line of business entirely
5. No need of expensive specialist resources such as claims handlers, lawyers to manage the liabilities.
6. It can improve the credit rating for the insurer.
7. In case the new insurer/reinsurer defaults, no liability on the existing insurer.

Disadvantages

1. Assets may need to be realized to pass across the reserves to the accepting insurer, this is important specially, if there are mismatching of reserves or liquidity concerns.
2. Default by new insurer/reinsurer may damage the reputation of the original insurer.
3. The transfer may require the buy-in of reinsurers where there are existing reinsurance arrangements covering the portfolio.
4. There would be associated cost for the risk transfer, which will depend on the current risk appetite of the market. As mentioned, the market is not profitable at the moment, the cost might be huge.
5. The process of finding and availability of insurer/reinsurer at an agreed term would be very difficult. It would be time consuming and in case it is available, it may be costly.
6. Approvals from regulatory authorities would also need to be obtained which might again be a difficult and time-consuming process.
7. If the transferred value of reserves uses a discount rate that is too low then there might be loss for insurer as he would have transferred the liabilities at a higher amount than he should transfer at.
8. Company may lose business in other lines as they may prefer to buy other products from accepting insurer
9. Some of the intermediary may leave company due to unavailability of this product

(Max 6)

Conclusion

The company should carefully consider all the above points before deciding the right option. If the company finds the right price and the premium payable for transfer of reserves is appropriate, the company should transfer the risks. Else, the company can wait and considers how the market is behaving and accordingly can decide the future course of action.

(1)

(11)

b) Other method of transferring reserve development risks in addition to mentioned above

Adverse Development Cover

It is a reinsurance arrangement whereby a reinsurer agrees, in return for a premium, to cover the ultimate settled amount of a particular line of business above a certain pre-agreed amount.

Feature of Adverse development cover

1. It protects the cedant from significant reserve deterioration.
2. This caps the liability and protects the balance sheet from any further development on existing losses and from future losses in respect of old business.
3. There could be an upper limit on the claims the insurer/reinsurer would accept.
4. Claims handling expenses and personnel cost are still borne by original insurer.
5. All the investment returns generated are earned by the insurer.
6. There is no transfer of reserves from the insurer to reinsurer.

7. Only a premium is paid for insurer/reinsurer taking on the responsibility of reserves beyond a specified limit.
8. Insurer is exposed to credit risk of reinsurer.

(Max 5)
[16 Marks]

Solution 3:

- 1) The company's outwards reinsurance premium is 48% of its inwards premium.
- 2) This would generally be considered to be a high proportion.
- 3) It may indicate that the company writes high-risk business that needs a lot of protection, or possibly that it obtains a lot of business from other companies with a reciprocal arrangement.
- 4) The company may also be a new company since new companies generally cede higher to reduce risk.
- 5) It is not likely to need surplus relief, as it has a high level of free capital.
- 6) The gross loss ratio is 144%, but the net is only 104%.
- 7) This may indicate the clever use of a soft reinsurance market/stop loss/excess of loss treaty, or possibly one or two major losses occurred or developed during 2023-24 that meant that the reinsurance programme was extensively used.
- 8) Also, it has greatly reduced its commission expense from 20% on gross basis to 10% on net basis. This may again indicate a clever use of reinsurance treaties/market, where RI income is large to reduce overall commission outgo on net basis.
- 9) Technical reserves are more than 7 years' premium and major component being IBNR indicating that company has a good amount of exposure in long tail line of business
- 10) Unless the company's business has reduced greatly, and we know that it has not in the past year at least, this suggests that the company's business is very long tail.
- 11) This is how the company can be profitable: it generates enough investment income to offset the effects of a loss ratio of 104%, commission of 10%, and expenses of management of 23%, a combined ratio of 137%.
- 12) Capital is 1500, about 200% of net premium. This would normally be considered a very high ratio, but with such a long tail account, and the possibility of needing to strengthen reserves, it is probably merely prudent.
- 13) Assuming the reserves are not discounted. the effective solvency position is even stronger than this.
- 14) The assets show a larger proportion of longer duration fixed interest securities. This is probably related to the long tail nature of the account.
- 15) Presumably this reflects a large portion of liabilities being to some extent linked to inflation.
- 16) The return on capital is 16%. This is healthy, but not extremely high considering high risk business, and would be lower still if reserves were discounted.
- 17) The company's assets yielded about 8.75% during 2023-24.

[Max 10 Marks]

Solution 4:

- a) Impact on overall valuation results at next exercise:
 - 1) Increased uncertainty in results
 - 2) Difficult in selecting assumptions as historical triangulated experience becomes irrelevant
 - 3) Increase in risk margin while calculating IBNR
 - 4) Lack of comparability of results between reviews,

- 5) may be difficult to identify differences in underlying claims experience from change in claims settlement approaches
- 6) Change in assumptions for next exercise as previous selections may be invalid
- 7) more time may be required to make entirely new selections as cannot start with previous assumptions
- 8) Reduced investigation of claims, claims department may pay out on more claims including fraudulent or erroneous claims if they are not verifying claims
- 9) Cost of claims handling may increase if more staff or more senior staff is employed
- 10) Reduction in tail - expect the estimate of unpaid claims on older accident years to reduce
- 11) Claims may be settled prematurely which may cause an increase in reopened claims if claims settlement is disputed
- 12) Expected change in average claim costs as faster claims settlement may lead to higher or lower costs per claim
- 13) Overall results may reduce: If ongoing claims management/monitoring was expensive then removing this cost may reduce overall results
- 14) If extensive claims investigation added considerably to costs, then settling all claims quickly may be a cheaper alternative.
- 15) Reduction in time of settlement reduces claim inflation which might reduce ACPC.

(Max 6)

b) Suggestions for further information to obtain:

- 1) What are the intended changes? What were the processes earlier and how it is different?
- 2) It would be helpful to understand how the claims management team are intending to achieve this. For example, a case study of a claim prior and post implementation of the strategy. Then the actuary could estimate the potential impact on claims which are currently reported but unpaid, and then on claims going forward.
- 3) How has the estimated revised term been calculated? What data is it based on?
- 4) How reliable is the estimated change.
- 5) How is this going to be monitored?
- 6) What information will the actuary have at the next review, or like to have at the next review in order to perform valuation.
- 7) How much of the most recent data reflects the change.
- 8) How much percentage of claim are reopened now and earlier?

(Max 2)

c) Options for valuation model:

- 1) Adopt entirely different valuation model although this makes comparability to previous valuations and measuring the impact of the changed strategy even more problematic, and loss of previously gained knowledge. May also be reasons why the ACPC model is preferable for comparability with other classes and existing monitoring tools. However, the ACPC model does not appear to cope well with the large claims in any case.
- 2) Could split ACPC by claim type to give greater information about how the strategy varies by claim type particularly if this is relevant to how the strategy is being implemented but there could be possibly too much volatility in the triangle to be useful/reliable.
- 3) Could split ACPC model between claims reported prior to strategy being implemented and claims after strategy – particularly if the strategy only applies to newly reported

claims. This does mean that there are two sets of factors and two triangles (as triangles are by accident year not report year) which splits the information.

(Max 2)

d)

- 1) Change in types of claims covered:
- 2) For example, changing way claims are paid i.e. from indemnity to benefit policy
- 3) For example, changing in cover to change claims types
- 4) Change in the nature of coverage
- 5) For example - only claims reported during the year are covered, rather than losses occurring during the year
- 6) Medical advances changing nature of claims such as treatments earlier requiring hospitalization may be treated using modern techniques in daycare
- 7) Limit on duration of cover for certain types of claims for e.g. reducing Temporary Total Disability claim coverage from maximum 100 weeks to maximum 20 weeks.
- 8) Change in excess – More or a smaller number of claims will be reported/settled
- 9) Reduced complexity in claims
- 10) legislative changes
- 11) Limitation in claim reporting time limits
- 12) Changes in business volume throughout the year
- 13) Increase in claims handling staff
- 14) Improved claims handling processes – streamlining or training of personnel
- 15) Change in business mix
- 16) Environmental change

(Max 6)

[16 Marks]

Solution 5:

a)

- 1) By exploring combinations of randomly-generated values for the key parameters, we can explore scenarios that we might otherwise not have imagined to be important
- 2) For example, two variables in the model may have a strong interaction that is not obvious
- 3) deterministic stresses might not have considered them both together
- 4) Similarly, a stochastic model makes it easier to explore 'ripple effects', that is, the knock-on consequences of the crystallisation of a risk event.
- 5) By estimating the full distribution of the aggregate result from the model, we can apply a number of different risk measures to the same set of final output
- 6) This may be important in sharing results with stakeholders who have different risk appetites or concerns
- 7) Even if we have calibrated individual stresses to represent a certain degree of extremity in their individual values, it can be very difficult to combine those stresses in a deterministic model
- 8) whereas a stochastic model allows us to be sure that the final result represents a sufficiently extreme combination of circumstances
- 9) A stochastic approach - in theory at least - explores all possible combinations and can rank these against the chosen risk measure
- 10) We can derive a probability distribution from the outcomes of a stochastic model and calculate confidence levels, if required
- 11) While the assumptions are subject to parameter error, the model is at least explicit about the assumptions being made
- 12) Every company writes different levels of each line of business and new emerging large risks have made stochastic model more important, e.g. increasing exposure to catastrophes, huge liability risks and upcoming surety bond risks.

- 13) The regulator may be interested in understanding the ranges around the point estimate of the model
- 14) New accounting standards might require a risk margin which might be best calculated on a stochastic basis
- 15) Allows for a better understanding of the uncertainty, depending upon the lines of business the company might be writing
- 16) Stochastic model will allow a better understanding/modelling of any guarantees embedded within the products/prices etc
- 17) Could result in lower capital requirements
- 18) Might be required for reinsurance pricing

(Max 6)

b) Insurance risk is made of underwriting risk (risks yet to be written/earned) and reserving risk (risk already written/earned).

Underwriting Risk

- 1) Gross Underwriting risk will be based on a realistic assessment of the insurer's business plan.
- 2) In case the data on personal lines building insurance is not relevant, the insurer must ensure that its assumptions are consistent with market experience, after adjusting for any differences such as the target market being small buildings.
- 3) The underwriting risk is the difference between the actual underwriting result, and that expected on a realistic basis.
- 4) One way to apply a realistic basis would be to exclude any profit expected on the business from this calculation and deduct it from the capital requirement as a separate item.
- 5) The insurer will need to consider how to divide its business for modelling purposes, e.g. by size of building, level of cover and territory.
- 6) This may be constrained by the lack of relevant internal data, leading to a lower level of grouping than is ideal.
- 7) In case data is relevant, consideration should be given to subdivisions such that it is not too small.
- 8) Most claims will be attritional claims, although there may be some large claims, depending on the insurer's typical retention for this line of business.
- 9) If the insurer limits the size of the risk taken on, then it should be able to limit the number of large claims.
- 10) In this case, it may be more practical to model loss ratios rather than separately model individual large claims, at least until relevant internal data becomes available.
- 11) Attritional claims should be modelled either using:
 - A lognormal distribution, if enough data is available to set appropriate parameters
 - A normal distribution, if a low enough level of variability is expected over claim sizes
 - Loss ratios
- 12) The insurer will probably decide to set a maximum size for an attritional claim as it is typical reinsurance retention for the class.
- 13) Any large claims should be modelled separately, and reinsurance recoveries allowed for on a frequency-severity basis.
- 14) Poisson and Negative binomial distribution are often used for frequency.
- 15) For severity, sampling from revalued past claims may be used. A heavily skewed distribution such as pareto would normally be appropriate.

- 16) For both attritional and large claims, it will be necessary to consider the if there are any types of claims that are not present in the historical claims data used for parameterising the distributions.
- 17) This can be done by increasing the standard deviation of the distributions beyond that derived purely from historical data.
- 18) Catastrophe claims are possible for this class of business, e.g. natural disaster and terrorist attacks
- 19) Catastrophe claims should be modelled using simulation, based on market catastrophe models.
- 20) However, other (human-made) catastrophes will have to be modelled on a bespoke basis, since commercial models may not be available.
- 21) Future latent claims are unlikely for this class, although some broad allowance may be made in respect of this.
- 22) If the insurer is using a multi-year model, then the underwriting risks from different policy years must be adjusted to allow for correlations.

Reserving Risk

- 23) Gross Reserving risk is in respect of business already written/earned.
- 24) This should be based on best estimate of reserves and necessary adjustments must be done if there are any significant reserve risk margin in the published reserves.
- 25) The capital impact of the reserving risk will be the difference between the eventual cost of settling the claims and the current reserves on best estimate basis.
- 26) The insurer will need to consider how to divide its business for modelling purposes, e.g. by size of building, level of cover and territory
- 27) The variability in the company's claim settling must be assessed by statistical techniques such as bootstrapping or Mack method.
- 28) It should be considered whether sufficient historical reserves shocks have occurred to indicate possible future variability
- 29) Latent claims shall be considered by a stress test as this cant be modelled with precision.

Net Insurance risk

- 30) It should be calculated by deducting the reinsurance recoveries rather than using net to gross ratios as there can be exhaustion of covers or disputes.
- 31) Diversification effects must be allowed for between underwriting risks and reserving risks.
- 32) Underwriting cycle must be considered as the model should reflect any expected movement of rates, terms & conditions, premium volumes and claims.
- 33) A margin should be included in the variability assumptions to allow for less than full credibility of historical data
- 34) Since in this scenario, the reinsurance is on loss occurring basis, the insurer will not have yet purchased reinsurance cover, the model should assume that reinsurance is not bought or allow for likely cost of purchasing the reinsurance.

(Max 12)

c)

- 1) Understanding the cover provided
- 2) Understanding the level of risks
- 3) The range of limits and sub-limits on offer
- 4) The limit and attachment profile of the portfolio
- 5) How underwriters select risk and decide upon the terms to offer in each case
- 6) Underwriting guidelines
- 7) The line size profile of the portfolio
- 8) Risk appetite limit set by management

- 9) The consistency and efficiency of operating and underwriting controls
- 10) Changes in legal, regulatory and business operating environment
- 11) Understanding the reinsurance programmes such as losses- occurring or risk attaching, reinstatements
- 12) Availability of data
- 13) Issues with internal data
- 14) Issues with other sources of data
- 15) Level of detail in a capital model
- 16) Sensitivity testing
- 17) Communicating uncertainty
- 18) Comparing the results
- 19) Use of capital model (regulatory or internal)
- 20) Consistency of assumptions
- 21) Documentation

(Max 6)
[24 Marks]

Solution 6:

The possible risks & uncertainties that the company may be facing at this point of time are

- 1) Random Error / Process Error: This covers the fundamental uncertainty (the insurable risk) that is insured.
E.g. – Number & amount of claims
Claim payment patterns
Changing development patterns
New Claim handling procedures (e.g. moving to online reporting) – Since it's a old company and the environment is changing, this may distort reporting patterns.
- 2) Parameter Error: This covers the uncertainty arising from the estimation of the parameters used in the model.
E.g.- Uncertainty arising from data used (likely in this case due to legacy systems)
Changes in case estimate reserving philosophy
Large & exceptional claim
Claim inflation not as expected
New Distribution channel
Planned/Unplanned changes in mix
- 3) Data Errors from systems & procedures: This covers the uncertainty arising from the way past claims are reported and stored.
E.g. – Legacy systems as in this case the company is 20 years old
Multiple data system
- 4) Model Error: This covers the uncertainty in the selection of the model and how it is used.
E.g.- Programming/Coding Error
Simulation Error
Model design being incorrect
(Since the company is using a statistical software, sometimes lack of proper understanding may result into it being used as a black box approach)
- 5) Adjustment factors: It is necessary to make a number of assumptions about how to adjust observed data to reflect the future experience, each of which is uncertain.
E.g. – Adjustment of ALAE/ULAE factors while setting up claim reserves
assumption of persistency ratio while setting renewal budget
- 6) Market Conditions: Insurance is not written in isolation. Legal, Regulatory and competitive effects can all influence the risk and uncertainty.
E.g. – inflation

New claim types
competition

- 7) Portfolio movements: Unexpected changes in the volume and mix of business may cause an insurer uncertainty.
E.g. – unexpected changes in volumes may lead to increase/decrease in ULAE if it is used as a % of claim amounts
May affect the level of commissions and expenses

(Max 7)

One way to remove uncertainty is to use a risk loading in addition to best estimate reserves, so that the resultant reserve levels are prudent.

(1)

[8 Marks]

Solution 7:

a)

AY/DY	12	24	36	48	60	72	84	96	EP
2017	200	17,500	38,500	44,500	48,800	52,700	54,900	54,900	96,200
2018	600	18,300	41,100	49,500	54,000	56,900	57,300		93,600
2019	600	18,700	50,900	60,900	64,500	66,700			1,05,400
2020	300	15,300	52,700	64,100	69,700				1,20,100
2021	400	15,200	55,800	65,200					1,36,000
2022	400	14,800	50,700						1,41,000
2023	400	16,500							1,44,800
2024	200								1,68,900

Weighted Average - All AY	40.10	2.90	1.19	1.08	1.05	1.02	1.00	1.00	
CDF	161.61	4.03	1.39	1.17	1.08	1.02	1.00	1.00	

AY	Paid	CDF	Ultimate - CL	Earned Premium	LR - CL
2024	200	161.61	32,322	1,68,900	19%
2023	16,500	4.03	66,492	1,44,800	46%
2022	50,700	1.39	70,385	1,41,000	50%
2021	65,200	1.17	76,119	1,36,000	56%
2020	69,700	1.08	75,192	1,20,100	63%
2019	66,700	1.02	68,282	1,05,400	65%
2018	57,300	1.00	57,300	93,600	61%
2017	54,900	1.00	54,900	96,200	57%

Assumptions

1. The first accident year is fully developed

2. The future development will follow the prior year's development
3. The past inflation present in data will be repeated in future.

(7)

b) Issues with the above method

1. One of large losses distorts the development patterns
2. Calendar year effect due to claims handling procedures
3. Results can be distorted by unusual experience.
4. Limited use for recent year cohorts, particularly for long tailed lines as in this case
5. No fully developed cohorts available in case of long tailed lines
6. In case of lines of business, where there are no claims or very few claims or only few large claims such as Liability line of business, claim projections cannot be done using this method.
7. It ignores the number of claims altogether.
8. Past inflation may not necessarily be the same in the future.

(Max 2)

c) Formula For BF Mehtod

$$\text{Ultimate Claims} = \text{Paid Claims} + ((1 - Z) * \text{ELR} * \text{EP})$$

Where Z is the development %

ELR is the expected loss ratio

EP is the Earned Premium

AY	Cumulative paid	CDF	Ultimate - CL	Earned Premium	LR - CL	Development Factor	Ultimate Claims - BF	LR - BF Method
2024	200	161.61	32,322	1,68,900	19%	1%	100,913	60%
2023	16,500	4.03	66,492	1,44,800	46%	25%	81,821	57%
2022	50,700	1.39	70,385	1,41,000	50%	72%	74,360	53%
2021	65,200	1.17	76,119	1,36,000	56%	86%	76,905	57%
2020	69,700	1.08	75,192	1,20,100	63%	93%	74,963	62%
2019	66,700	1.02	68,282	1,05,400	65%	98%	68,165	65%
2018	57,300	1.00	57,300	93,600	61%	100%	57,300	61%
2017	54,900	1.00	54,900	96,200	57%	100%	54,900	57%

(4)

d) BF Method is highly dependent on priori loss ratio / Expected loss ratio which is largely based on judgement.

(1)

e) Cape Cod method can be used instead of BF Method, as the expected claims ratio is obtained from the reported claims experience instead of independent and often judgemental selection in the BF Method.

(1)

[15 Marks]**Solution 8:**

- 1) The Topline i.e. GWP would reduce drastically if the company has higher exposure in multi-year book.
- 2) Commission may still need to be paid on the total money received by the company.
- 3) Thus, Commission to GWP ratio may increase disproportionately.
- 4) The Expenses to GWP/NWP ratio will also increase due to the mentioned change.
- 5) The Combined Ratio as defined by the regulator may seem to be very high, which may not actually show the correct picture.

- 6) The company may breach its total expenses to GWP proportion due to the accounting change.
- 7) The Solvency ratio may also get impacted due to change in accounted GWP.
- 8) This can be perceived wrongly by the public and may lead to decrease in share price in case of listed companies.
- 9) There would be need to change the IT systems and may be a time-consuming exercise.
- 10) The company may have to revisit its Budgeting exercise and re-draw its future plans.
- 11) Accounting for Reinsurance treaties may become difficult as Reinsurance treaties written on Risk attaching during (RAD) basis may have risks coming from different UW years during a particular year.
- 12) The earned premium for the company may fall sharply if it has high exposure in multi-year policies since it has opted for the option to earn 50% of written premium in the same accounting year.
- 13) In case the company haven't opted for the option, its earned premium wouldn't have change.
- 14) The company profitability would also be affected as earned premium might reduce.
- 15) The company may decide to reduce its exposure in multi-year policies and thus may loose on investment income.
- 16) In case, multi-year policies have different claims experience as compared to one- year policies, and company changes its proportion of business, it will lead to change in overall claims ratio and thus underwriting profit.
- 17) The above issues could be solved to some extent if acquisitions costs are allowed to be deferred.
- 18) In case the company delays payment of commission to its intermediaries, it may impact the confidence and trust of intermediaries and thus may loose business opportunities.

[Max 6 Marks]
